Janhvi Pradip Bankar

Computer Science Undergraduate

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Professional Summary

Computer Science undergraduate with experience in AI/ML applications, full-stack development, computer vision, predictive analytics and Retrieval-Augmented Generation (RAG) systems.

Education

B.Tech (CSE) – Rajarambapu Institute of Technology, Islampur, Maharashtra **CPI: 8.37 - 2027**

HSC – New English School Jr. College, Sangola, Maharashtra **81.83% - 2023**

SSC – New English School, Sangola, Maharashtra **97.20% - 2021**

Technical Skills

Programming Languages: C, Core Java, JavaScript
Frontend & Mobile: React.js, Flutter
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs
AI/ML & Computer Vision: Scikit-learn, YOLO, CNN, OpenCV, MediaPipe, SHAP
Generative AI: RAG, LangChain, Embeddings, Vector Search
Databases: MySQL, MongoDB Atlas, Firebase Firestore
Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab
Core Areas: OOP, DBMS

Projects

AI-Based Bharatanatyam Learning App

- Developed a learning application featuring real-time Bharatanatyam mudra recognition and practice assistance.
- Implemented gesture recognition using YOLO/CNN, OpenCV and MediaPipe, with a custom dataset and Flutter application integration.
- Designed real-time feedback to assess dance gesture accuracy.

MigraineGuardian - AI Migraine Risk Prediction System

- Developing a daily migraine risk prediction system using lifestyle indicators, including sleep, stress, hydration, mood and screen time.
- Built a FastAPI-based ML service and evaluated experimental prediction models incorporating weather features.
- Integrated Firebase Firestore for user check-ins and weather records, with React and Node.js/Express for the application backend.
- Implemented SHAP-based explanations and integrated a RAG chatbot for document-grounded healthcare information.

RAG Chatbot - MigraineGuardian Module

- Developed a RAG pipeline with LangChain-based offline document loading, chunking, embeddings and vector retrieval.
- Integrated MongoDB Atlas for vector storage and persistent conversational history, with the six most recent messages used to provide context for response generation.
- Integrated Groq LLM inference and Google AI API services with a Node.js/Express backend and React frontend.

Hospital Management System - MERN Stack

- Developed a full-stack application for patient registration and appointment scheduling.
- Implemented role-based authentication and optimized MongoDB queries to improve data retrieval performance.
- Designed REST APIs for secure backend communication.

Achievements & Responsibilities

Winner - Smart India Hackathon (SIH) 2025

UI/UX Head - OysterCode Club

Social Media Head - Cryptex Club

Contributor - Social Shuffle Initiative

Certifications

- NPTEL - Machine Learning (IIT Madras, 12 Weeks)
- NPTEL - Java Programming (IIT Kharagpur, 12 Weeks)
- NPTEL - Database Management Systems (IIT Kharagpur, 8 Weeks)
- Google - iOS App Development Certification
- Infosys Springboard - Theory of Computation
- Infosys Springboard - Advanced HTML5 & CSS3